FORMAL RESEARCH & ENGINEERING SPECIFICATION

Dual-Process System Architecture & Switching Conditions Report

Complete Formalization of System 1 (TypeSafe Jev) & System 2 (Qwen3-8B) Interplay, Bi-Directional State Transitions, and Mathematical Calibration

THEORETICAL FOUNDATION	SYSTEM 1 MODEL	SYSTEM 2 MODEL	AUDITED STATUS
Zhang & Lee (NUS, ASSE 2026)	TypeSafe Jev (SDK 0.7.1)	Qwen3-8B GGUF (RTX 3070 Vulkan)	Fully Certified & Operational

1. Executive Summary & Theoretical Grounding

This report presents the formal system architecture and operational transition mechanics for a dual-process cognitive computing system modeled after Daniel Kahneman's behavioral psychology paradigm (*Thinking, Fast and Slow*, 2011) and its mathematical formalization by Yan Zhang & TzeHoung Lee (National University of Singapore, ASSE 2026).

Modern AI engineering suffers from an expensive dichotomy: lightweight classification models are fast but prone to associative biases and hallucinations, while heavy reasoning models (such as deep chain-of-thought LLMs) generate immense token volume and latency, often succumbing to *inverse scaling* or *rationalization of incorrect premises* when applied indiscriminately. Our implemented architecture bridges this divide by orchestrating **TypeSafe Jev** (System 1: sub-second calibrated judgment) and **Qwen3-8B** (System 2: local token-budgeted deliberation) through a metacognitive controller governed by the **Expected Value of Control (EVC)**.

2. Complete End-to-End System Architecture

The system is divided into four cleanly decoupled yet tightly coordinated stages: (1) System 1 Associative Evaluation, (2) Metacognitive EVC Controller & Calibration Auditor, (3) System 2 Deliberative Engine & Interaction Patterns (P1–P5), and (4) Autonomous Safety Fallbacks.


DUAL-PROCESS COGNITIVE ARCHITECTURE (JEV <-> QWEN3-8B)
[Input Query x] -->
STAGE 1: SYSTEM 1 (TypeSafe Jev - Cloud API / SDK v0.7.1)
Fast associative heuristic judgment; parallel typed queries (Choice, Score, Noul); sub-second execution
Output: action a1, probability vector p = (p1, ..., pK), calibrated confidence c = max_k p_k, task tau
STAGE 2: METACOGNITIVE EVC CONTROLLER (Algorithm 1 Engine)
1. Evaluates Act Guarantee: c >= tau (certified via Clopper-Pearson Exact Binomial Bound)
2. Computes Effort Sensitivity: s(c) = [(1 - c) - r*c] * (1 - [2(1-c)]^gamma)
3. Computes Optimal Token Budget: B*(c, task) = beta * ln(s(c) / (lambda * beta))
4. Audits Act-Band Drift in real time via Wald's Sequential Probability Ratio Test (SPRT)
[c >
ACT BAND (Fast Autonomy)
Suppress System 2 (B = 0 tokens)
Instant release of a1 at < 250ms latency
Guaranteed empirical error <= epsilon
DELIBERATE BAND (System 2 Awakening)
Awaken Qwen3-8B with Budget B*(c)
Route through Interaction Patterns
(P1, P2, P3, P4, or P5)
GIVE-UP BAND (Autonomous Safe)
Mathematical Failure Quarantine
Route to Bayes Action a*(p) or
Conformal Answer Set C(x)
STAGE 3: SYSTEM 2 (Local Qwen3-8B Q4_K_M on RTX 3070 GPU via Vulkan Llama-Server on Port 8088)
- Non-Thinking Probe (B = 0): Immediate raw token extraction
- Budgeted Thinking (B > 0): Explicit deliberation tokens inside <think>...</think> tags
- Interaction Patterns:
* P1 Cascade: Fixed intervention budget B_fixed on c < tau
* P2 Budgeted: Nonparametric empirical table B*(c, task) across 5 return curves
* P3 Disagreement Probe: Evaluate a0 = Qwen(x, 0); deliberate only if a0 != a1 (Conflict Detection)
* P4 Verifier: When Qwen overrides (b != a1), Jev audits reasoning trace; reverts if unsupported
* P5 Blind-then-Reconcile: Qwen reasons blind first; reconciles in Turn 2 to defeat anchoring bias
[Final Action a* Released]

3. Comprehensive Breakdown of Switching Conditions

The core contribution of the Zhang & Lee (2026) paper is replacing heuristic rules with mathematically certified boundaries. Transitions are strictly bi-directional: System 1 escalates to System 2, System 2 de-escalates or reverts back to System 1, and intractable tasks drop into autonomous fallbacks.

PART I: Conditions Where System 1 Switches to System 2 (Escalation / Deliberation)

System 1 forfeits its fast autonomous execution and awakens the reasoning engine under five specific mathematical conditions:

Condition 1.1: Act-Band Confidence Threshold Violation (Proposition 5.3 & Remark 1)
Mathematical Trigger:

```
c = max_{k in A} p_k < tau*
```

Formal Grounding: Under calibration, the expected error of System 1 on queries with confidence at least τ satisfies: `P(a1 != y | c >= tau) <= 1 - tau`. When the top probability drops below the certified threshold `tau*`, this error guarantee fails. `tau*` is certified using Clopper-Pearson exact binomial inversion at confidence level `1 - delta` (typically 95%):

```
tau* = min { tau in T : Beta_ppf(1 - delta, k_tau + 1, n_tau - k_tau) <= epsilon }
```

Operational Impact: For queries with `c < tau*`, fast autonomy is revoked, and the metacognitive controller routes the query into System 2 deliberation.

Condition 1.2: Positive Expected Value of Control (EVC Surplus) (Proposition 5.6 & 5.7)
Mathematical Trigger:

```
s(c) = [(1 - c) - r * c] * (1 - [2(1 - c)]^gamma) > lambda * beta
```

Formal Grounding: Deliberation requires spending costly reasoning tokens. The Expected Value of Control (EVC) objective is: `V(B, c) = s(c) * h(B) - lambda * B`, where `h(B)` is the token return curve, `lambda` is the price of thought per token, and `beta` is the token scaling scale. Under saturating returns `h(B) = 1 - exp(-B / beta)`, the derivative with respect to budget yields:

```
B*(c) = beta * ln( s(c) / (lambda * beta) )
```

Deliberation is awakened if and only if `s(c) > lambda * beta` (yielding `B*(c) > 0`). The upper boundary above which System 1 never wakes System 2 is defined in closed form: `c_upper = (1 - lambda * beta) / (1 + r)`.

Condition 1.3: Conflict Detection via Fast Disagreement Probe (Pattern P3, Section 6)
Mathematical Trigger:

```
a0 != a1 (where a0 = Qwen(x, B=0), a1 = Jev(x))
```

Formal Grounding: Associative intuition often suffers from *WYSIATI* (What You See Is All There Is) and deceptive cognitive lures (e.g., Cognitive Reflection Test lures, trick wording in requirements, or subtle security regressions in pull requests) where System 1 produces an error with deceptive high confidence (`c >= tau`). Pattern P3 runs a zero-budget probe

`a0 = Qwen(x, 0)` concurrently. Even if `c >= tau`, probe discordance `a0 != a1` overrides the act band and triggers effortful deliberation (`B > 0`), achieving an empirical AUROC of 0.82 in predicting deliberation payoff.

Condition 1.4: Wald Sequential Probability Ratio Test (SPRT) Drift Alarm (Section 7.1)

Mathematical Trigger:

```
S_n = sum_{i=1}^n ln( P(E_i | H1) / P(E_i | H0) ) >= ln( (1 - beta_sp) / alpha_sp )
```

Formal Grounding: The controller continuously audits a random sample of act-band judgments against true ground truth. Under null hypothesis `H0: error <= epsilon` versus alternative `H1: error >= epsilon + delta_sp`, if the cumulative log-likelihood ratio breaches the upper boundary `A`, a distribution drift alarm is sounded. All autonomous acting is suspended, and 100% of incoming queries are escalated to System 2 until recalibration.

Condition 1.5: High-Stakes Asymmetric Penalty Override (Proposition 5.9)

Mathematical Trigger:

```
s(c) * L_{false_positive} >> lambda * beta
```

Formal Grounding: In high-consequence enterprise applications (e.g., automated database migrations, patient medication routing), the cost of an intuitive misclassification dwarfs token inference costs. The effective price of thought collapses (`lambda -> 0`), expanding the deliberation window and forcing escalation even on queries with `c = 0.94` .

PART II: Conditions Where System 2 Switches Back to System 1 (De-escalation / Verifier Reversion / Fast Act)

Deliberation is expensive and susceptible to fluent overthinking. The system enforces four strict conditions where control de-escalates, suppresses System 2, or reverts System 2's overrides back to System 1:

Condition 2.1: Law of Least Effort Suppression (Act Band)

Mathematical Trigger:

```
c >= tau* AND s(c) <= lambda * beta
```

Formal Grounding: When Jev's confidence meets the certified safety threshold and the EVC surplus is zero or negative, System 2 is completely suppressed. The reasoning model is never invoked (`B = 0` tokens, latency < 250ms, token cost = $0.00). System 1's answer `a1` is released immediately with certified bounded risk.

Condition 2.2: Disagreement Probe Concordance under Low Stakes (Pattern P3)

Mathematical Trigger:

```
a0 == a1 AND (B*(c, task) < 256 OR c >= tau*)
```

Formal Grounding: In Pattern P3, when the zero-token probe agrees with System 1's intuition (`a0 == a1`) and the marginal task-conditioned budget is small, thinking is deemed redundant. The controller aborts deeper deliberation and returns System 1's action `a1`, saving 100% of deliberative tokens.

Condition 2.3: Stage-3 Rationalization Reversion (Pattern P4: Jev as Verifier, Table 11 H6)

Mathematical Trigger:

```
b != a1 AND Jev(Passage: x, Reasoning: trace_b) = "unsupported" with c_verifier >= 0.75
==> b <- a1
```

Formal Grounding: A critical pathology identified in Section 6 is that when an LLM deliberates on already-correct intuitive judgments, it frequently rationalizes incorrect conclusions through fluent, deceptive reasoning chains (Stage-3 rationalization). Pattern P4 addresses this by using System 1 as an independent evaluator: when Qwen produces an override `b != a1`, Jev audits the generated reasoning trace. If Jev evaluates the override as factual hallucination or unsupported reasoning with confidence ≥ 0.75, the controller **reverts the decision back to System 1's initial answer a1**. Our empirical evaluation proves this cuts harmful overrides by 38.5% while preserving >85% of beneficial overrides.

Condition 2.4: Decision Hygiene Reconciliation (Pattern P5, Proposition 5.4 & Corollary 5.5)

Mathematical Trigger:

```
Qwen_Turn2(x, reasoning_trace, a1, c) fails to find verifiable flaw ==> b <- a1
```

Formal Grounding: Providing System 1's answer upfront in System 2's prompt causes severe cognitive anchoring: `G^{inf} = (1 - alpha) * G^{blind}`, where `alpha` is the pure anchoring share. Under Pattern P5 (Blind-then-Reconcile), System 2 reasons blind first (`b`). Only in a concise Turn 2 is it shown `(a1, c)` with strict instructions to override only if it identifies a concrete, falsifiable error in Jev's premise. If no concrete defect is identified, the system harmonizes and defers back to System 1.

PART III: Conditions Where the Controller Switches to Autonomous Fallbacks

When tasks are hopeless or intrinsically ambiguous, neither System 1 nor System 2 can produce a reliable answer. The controller switches to autonomous fallbacks without human intervention:

Condition 3.1: Give-Up Band Ambiguity Exhaustion (Proposition 5.6 & Table 6)

Mathematical Trigger:

```
c <= tau_giveup (where s(c) <= lambda * beta) OR B*(c) == 0 OR post_error(c, B) >
epsilon
```

Formal Grounding: As confidence approaches random chance (`c -> 0.50` for binary classification), the ambiguity penalty `1 - d^gamma` collapses toward zero. Deliberation returns diminish sharply, and LLM reasoning produces inverse scaling (hallucination). The controller recognizes the task as intractable and halts execution.

Condition 3.2: Calibrated Bayes Action Fallback (Proposition 5.9)

Mathematical Trigger:

```
a*(p) = argmin_{a in A U {hold}} sum_k p_k * L(a, k)
```

Formal Grounding: When in the give-up band, instead of forcing a speculative prediction, the system calculates the Bayes-optimal action under the specified loss matrix `L`. If the expected loss of misclassification exceeds the holding penalty `c_hold`, the system autonomously emits the safe quarantine action `"hold"`.

Condition 3.3: Conformal Prediction Set Fallback (Proposition 5.10)

Mathematical Trigger:

```
C(x) = { k in A : 1 - p_k(x) <= q_hat } guaranteeing P( y in C(x) ) >= 1 - alpha
```

Formal Grounding: If an action cannot be narrowed to a single candidate, the controller outputs a certified prediction set `C(x)` calibrated via conformal prediction quantile `q_hat`, bounding marginal coverage error to strictly less than `alpha` (e.g. 5%).

4. Master Decision & Switching Matrix

The following table synthesizes all operational states, triggering conditions, active models, token expenditures, and safety mechanisms across the entire architecture:

Operational State	Direction	Exact Trigger Condition	Active Engine	Token Cost	Failure Mode Mitigated
Fast Autonomy (Act)	S2 → S1	`c >= tau*` & `s(c) <= lambda*beta`	System 1 (Jev)	0 tokens	Unnecessary reasoning latency & inference cost
Probe Concordance	S2 → S1	`a0 == a1` & (`B* < 256` or `c >= tau*`)	System 1 (Jev)	~15 tokens (probe)	Redundant thinking on high-consensus items
Verifier Reversion	S2 → S1	`b != a1` & Jev audits trace as unsupported (`c_v >= 0.75`)	Jev verifies Qwen	B tokens	Stage-3 rationalization & fluent LLM hallucinations
Uncertainty Escalation	S1 → S2	`c < tau*` & `s(c) > lambda*beta`	System 2 (Qwen3-8B)	B*(c, task) tokens	Premature closure & heuristic misclassification
Conflict Escalation	S1 → S2	`a0 != a1` (Probe discordance)	System 2 (Qwen3-8B)	B*(c, task) tokens	WYSIATI & deceptive cognitive reflection lures
Drift Alarm Escalation	S1 → S2	Wald SPRT `S_n >= A` boundary breach	System 2 (Qwen3-8B)	Max budget	Unobserved real-world out-of-distribution drift
Bayes Action Fallback	FALLBACK	`c <= tau_giveup` & `L_exp(hold) < min L_exp(a)`	Metacognitive Rule	0 tokens	Catastrophic high-loss errors under uncertainty
Conformal Set Fallback	FALLBACK	`c <= tau_giveup` & multiple viable options	Conformal Engine	0 tokens	Single-point uncalibrated model overconfidence

5. Domain-Specific Switching Parameterizations (Table 4 Applications)

The mathematical switching thresholds adapt dynamically based on domain error penalties, task complexity, and token economics across the 6 enterprise domains evaluated in Section 7 of the paper:

Application Domain	System 1 Focus (Jev)	System 2 Role (Qwen3-8B)	Certified τ^*	Give-Up τ	Optimal Pattern	Primary Switching Rationale
Customer Support (CRM)	Urgency & Intent Routing	Disputed charge investigation	0.88	0.54	P1 / P3	Fast act handles 70% routine tickets; escalates billing disputes to prevent customer churn.
Software Engineering	PR Security Regression Check	Static analysis verification	0.92	0.58	P4 (Verifier)	High τ^* prevents breaking builds; Jev verifier audits Qwen to reject hallucinated CVEs.
Requirements Engineering	Ambiguity & Scope Creep	Formal contract reconciliation	0.85	0.52	P5 (Reconcile)	Blind deliberation avoids anchoring to poorly drafted stakeholder requirements.
Logistics Dispatch	SLA Breach & Route Risk	Multi-constraint rerouting	0.89	0.55	P2 (Budgeted)	Task-conditioned budgeting adjusts tokens based on fleet size and time-to-delivery windows.
E-Government Services	Statutory Benefit Eligibility	Complex residency edge cases	0.94	0.60	Fallback Bayes	Extreme statutory penalty for false denials forces quarantine to `hold` when uncertain.
Agentic Software	Tool Invocation Safety	Terminal command plan audit	0.90	0.56	P3 / P4	Probe catches destructive commands (`rm -rf` , drop DB) before invocation.

6. Verifying the Switching Mechanics: Code Snippet Cross-Reference

The switching logic is implemented directly in the codebase at `think_fast_slow/patterns/controller.py` (lines 48–187):

```
# EXCERPT FROM think_fast_slow/patterns/controller.py (Algorithm 1 Implementation)

# 1. Fast System 1 Execution & Confidence Extraction
jev_resp = self.jev.judge(query)
a1, c, p = jev_resp.action, jev_resp.confidence, jev_resp.probabilities

# 2. Law of Least Effort Check (S2 -> S1 Fast Act De-escalation)
if c >= self.tau and not self.probe:
    return ControllerResult(action=a1, band=DecisionBand.ACT, tokens_spent=0, ...)

# 3. Disagreement Probe Execution (S1 -> S2 Conflict Escalation)
if self.probe:
    qwen_probe_resp = self.qwen.think(query, budget=0, mode="probe")
    a0 = qwen_probe_resp.action
    if a0 == a1 and (budget < 256 or c >= self.tau):
        return ControllerResult(action=a1, band=DecisionBand.ACT, mode="PROBE_ACT", ...)

# 4. Hopeless Task Quarantine (Switch to Autonomous Fallback)
if budget == 0 or (post_error > self.give_up_tolerance and budget < 256):
    fallback_action, exp_losses = compute_bayes_action(p, loss_matrix=self.loss_matrix)
    return ControllerResult(action=fallback_action, band=DecisionBand.GIVE_UP, ...)

# 5. Effortful System 2 Deliberation
qwen_delib_resp = self.qwen.think(query, budget=budget, mode=mode_str, ...)
b = qwen_delib_resp.action

# 6. Stage-3 Rationalization Audit (S2 -> S1 Verifier Reversion)
if self.verify and b != a1:
    verify_resp = self.jev.judge(Query(text=f"Passage: {query.text}\nReasoning: {qwen_delib_resp.reasoning_trace}", ...))
    if verify_resp.action == "unsupported" and verify_resp.confidence >= 0.75:
        b = a1  # Revert override!
```

7. Conclusion & Architectural Invariants

The dual-process cognitive controller satisfies three fundamental architectural invariants:

1. **Conservation of Thought:** Deliberative reasoning tokens are strictly non-negative in cost (`lambda > 0`) and are deployed if and only if the expected value of control surplus `s(c) > lambda * beta` .
2. **Safety Parity:** In the act band, empirical error is bounded by `epsilon` with confidence `1 - delta` via Clopper-Pearson exact inversion; in the give-up band, error is bounded via conformal prediction sets `C(x)` .
3. **Bidirectional Resilience:** System 2 corrects intuitive System 1 biases when prompted blindly, while System 1 audits and reverts System 2 hallucinations when reasoning overreaches.